PACE: Probabilistic Adaptive Contention Control for Non-Primary Channel Access in IEEE 802.11bn

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***Abstract*—Modern Wi-Fi deployments increasingly suffer from spectrum underutilization as overlapping Basic Service Sets (OBSSs) compete for primary channel access. IEEE 802.11bn introduces Non-Primary Channel Access (NPCA) to exploit these underutilized intervals, yet the standard contention mechanism is fundamentally mismatched to NPCA's constrained transmission windows: backoff counters routinely exceed the remaining opportunity duration, wasting slots that could otherwise carry data. We propose PACE, a probabilistic adaptive contention control scheme that enables stations to collectively converge toward a theoretically optimal transmission probability without central coordination. PACE combines a distributed multiplicative feedback rule—copying the successful sender's probability on solo receptions—with a prospective self-exclusion mechanism that silences stations whose frame length exceeds the remaining window. We evaluate PACE through simulation spanning a wide range of station counts and window lengths, comparing against Binary Exponential Backoff and several competing adaptive schemes under both homogeneous and heterogeneous frame-length distributions. PACE consistently outperforms standard backoff in channel utilization and achieves Pareto-dominance in the throughput–fairness tradeoff. The advantage is most pronounced when the available window is scarce relative to the contending population, precisely the regime in which standard backoff wastes the largest share of transmission opportunities. These results identify the tight-window regime as the critical operating condition for NPCA and establish PACE as a practical design reference for adaptive contention policy in next-generation Wi-Fi networks.**

***Index Terms*—Non-Primary Channel Access, IEEE 802.11bn, Probabilistic Access Control, OBSS, Adaptive Contention, Wi-Fi**

I. Introduction

Spectrum efficiency in next-generation Wi-Fi networks depends critically on how effectively stations (STAs) exploit available channel time. In dense deployments, overlapping Basic Service Sets (OBSSs) routinely occupy the primary channel while adjacent channels remain idle, creating a persistent underutilization problem. IEEE 802.11bn addresses this gap by introducing Non-Primary Channel Access (NPCA), which allows STAs to transmit opportunistically on non-primary channels during ongoing OBSS intervals [?], [?].

NPCA, however, presents a contention challenge distinct from conventional channel access. Each NPCA opportunity has a finite duration bounded by the ongoing OBSS transmission, and competing STAs must coordinate their access within this constrained window. The Binary Exponential Backoff (BEB) mechanism was designed for open-ended contention: each collision doubles the backoff window, and the resulting delay frequently exceeds the remaining NPCA window duration. Affected STAs are effectively silenced for the rest of the interval, leaving the channel idle even when other STAs could have transmitted. In heterogeneous deployments where STAs carry frames of varying lengths, the problem is further compounded: STAs whose frame cannot fit in the remaining window continue to occupy contention slots despite having no viable transmission opportunity.

Consider a dense office environment where multiple STAs detect an OBSS transmission and attempt to use the secondary channel concurrently. Early collisions trigger exponential backoff growth; within a few rounds, a large fraction of STAs have accumulated backoff counts that exceed the remaining window duration. The channel spends the final slots idle, not because no STA wanted to transmit, but because the contention mechanism pushed every backoff counter beyond the opportunity horizon.

Existing work on NPCA has focused on characterizing aggregate throughput and delay under the standard switching rule, taking Binary Exponential Backoff within the non-primary channel as given [?], [?], [?]. The contention dynamics within the finite NPCA window—and the mismatch between BEB's infinite-horizon design and the window's bounded duration—have not been addressed. Adaptive probabilistic schemes designed for infinite-horizon neighbor discovery, such as the ALOHA-like Neighbor Discovery (AND) algorithm [?], reduce overhead by avoiding explicit coordination, but their stationary-population assumption makes them ill-suited to a window that depletes over time. BEB's infinite-horizon design and AND's open-loop phase structure both fail to track the time-varying optimal access probability as the window shrinks and the set of viable STAs changes (see Table I).

In this paper, we propose PACE, a Probabilistic Adaptive Contention control scheme for NPCA that enables STAs to collectively converge toward the theoretically optimal transmission probability for each moment within the window, without central coordination or explicit inter-STA signaling. PACE adapts the MIMD feedback rule of the Probabilistic Neighbor Discovery (PND) algorithm [?] to the finite-window NPCA context: STAs copy the successful sender's probability on solo

TABLE I: Comparison of Contention Control Schemes

Method	Mechanism	Objective	PPDU-aware	Signaling-free	NPCA target
AND [?]	Fixed-probability slotted ALOHA	Discover all neighbors in finite time without coordination	×	✓	×
PND [?]	Multiplicative feedback with solo-copy rule	Converge transmission probability to optimal without N	×	✓	×
Cali *et al.* [?]	p-persistent with contender estimation	Approach theoretical throughput limit in a distributed manner	×	✓	×
BLADE [?]	Hybrid increase / multiplicative decrease via access rate	Eliminate packet-delivery droughts for real-time Wi-Fi	×	✓	×
IYT [?]	Neighbor-ordered backoff interval assignment	Reduce collision probability and worst-case latency	×	✓	×
ACWB-DQN [?]	Deep Q-network contention window selection	Maximize throughput under varying network load	×	×	×
FC-MAPPO [?]	Fairness-constrained multi-agent policy optimization	Minimize collisions while ensuring inter-STA fairness	×	×	×
PACE (proposed)	**Multiplicative feedback with PPDU-aware self-exclusion**	**Maximize channel utilization in finite NPCA window**	✓	✓	✓

receptions, reduce on collision, and increase on idle slots, causing distributed dynamics to track the throughput-optimal policy. PACE further adds a PPDU-aware self-exclusion rule that silences STAs whose frame length exceeds the remaining window, eliminating a class of futile contention attempts that BEB would otherwise sustain.

The design of PACE is motivated by two requirements that are unique to NPCA's bounded-window structure: *(i)* the optimal transmission probability is time-varying and must track the depleting window rather than converge to a static equilibrium, and *(ii)* STAs whose frame length exceeds the remaining window must proactively self-exclude to prevent futile contention that wastes viable slots. The main contributions of this paper are as follows:

- We derive the throughput-optimal transmission probability for finite-window NPCA, showing that it must continuously track the remaining window duration and the number of still-viable competing STAs—a time-varying target that neither BEB nor existing infinite-horizon adaptive schemes can follow.
- We propose PACE, a fully distributed algorithm that addresses both requirements simultaneously: MIMD feedback drives the transmission probability toward the time-varying optimum *(i)*, while PPDU-aware self-exclusion eliminates futile contention by silencing STAs that cannot complete a frame within the remaining window *(ii)*, all without inter-STA signaling or knowledge of the total STA count.
- Through extensive simulation under homogeneous and heterogeneous frame-length distributions, we demonstrate that PACE achieves significantly higher window utilization than BEB, is Pareto-dominant in the throughput–fairness tradeoff, and maintains proportional fairness between native and visiting STAs.

The remainder of this paper is organized as follows. Section II reviews related work on NPCA and adaptive MAC protocols. Section III presents the system model. Section IV describes the PACE algorithm and its theoretical foundation. Section V presents simulation results, and Section VI concludes the paper.

II. Related Work

Table I summarizes the key distinctions among existing contention control schemes and the proposed PACE.

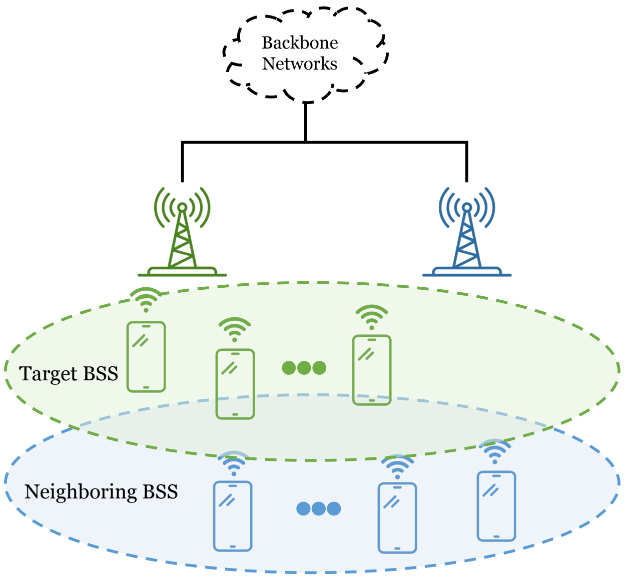
A. Non-Primary Channel Access in IEEE 802.11bn

IEEE 802.11bn introduces NPCA as a core mechanism for exploiting idle secondary channels during OBSS transmissions [?], [?]. Wei *et al.* [?] extended Bianchi's DCF model to a multi-channel setting, establishing analytically that NPCA always outperforms the legacy baseline; a follow-up study [?] incorporated switching overhead and proposed a threshold policy to toggle between NPCA and legacy modes based on primary channel occupancy. Bellalta *et al.* [?] developed a Continuous-Time Markov Chain model for a multi-BSS topology, identifying a secondary-channel contention tradeoff that can limit NPCA gains when the secondary channel is shared. Lee and Song [?] further showed that PPDU frame duration has a nonlinear effect on inter-BSS coexistence, and Song [?] applied deep reinforcement learning to the NPCA channel-activation decision.

Across these studies, however, the contention mechanism within the NPCA window is held fixed at Binary Exponential Backoff. How STAs should adapt their access probability as the window depletes and viable STAs drop out has not been studied.

B. Adaptive Contention Control in Wireless LANs

Bianchi [?] established that BEB operates below the theoretical throughput ceiling, and Cali *et al.* [?] showed that p-persistent access with dynamic contender estimation can approach this limit in a distributed manner. Subsequent work has diversified the adaptation mechanisms: BLADE [?] uses a HIMD policy driven by a universally observable channel signal; Wilhelmi *et al.* [?] impose structured transmission order via inter-BSS activity awareness; and learning-based approaches optimize the contention window through deep Q-networks [?] or fairness-constrained multi-agent policy optimization [?].



	Channel #1	Channel #2
Target BSS	BSS Primary	NPCA Primary
Neighboring BSS		BSS Primary
Overlapping BSS	OBSS Tx	

Fig. 1: Two-BSS topology and channel assignment.

Despite this breadth, all existing schemes assume STAs contend over an open-ended horizon and target steady-state convergence. NPCA's bounded window demands a fundamentally different approach: the optimal transmission probability must evolve as the window shrinks and the viable STA population changes. The MIMD feedback rule of PND [**?**] is the closest mechanism in spirit, but was designed for infinite-horizon neighbor discovery and provides no accounting for bounded window duration or frame-length viability.

III. System Model

A. Network and Channel Model

Fig. 1 depicts the two-BSS topology and channel assignment assumed throughout this work. The target BSS, whose NPCA operation is our focus, comprises an Access Point (AP) and a set $\mathcal{N} = \{1, 2, \ldots, N\}$ of associated non-AP STAs and they can be considered as visitors to the neighboring BSS. The visitors normally contend on their BSS primary channel, Channel #1 in Fig. 1, which is the main channel used by the target BSS for its own transmissions.

Channel #1 is also used by an overlapping BSS (OBSS) operating on the same channel, whose transmissions intermittently render it busy. In standards prior to IEEE 802.11bn, a STA could not initiate a transmission whenever its designated primary channel was busy, even if the secondary channel was idle. In IEEE 802.11bn, however, the AP can designate a separate channel as the NPCA primary channel, used by the visitors during NPCA operation, named as Channel #2 in Fig. 1. We use the qualifier "BSS" to distinguish the original

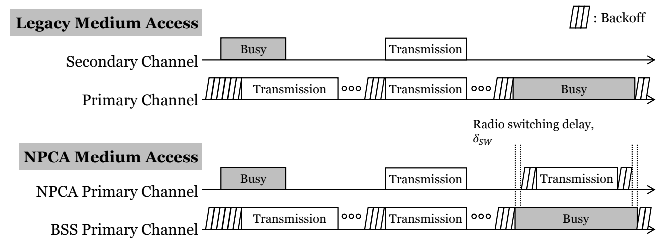


Fig. 2: The medium access process comparison between legacy and NPCA.

primary channel from the NPCA primary channel, consistent with P802.11bn Draft terminology [**?**].

B. Legacy versus NPCA Medium Access

Let us elaborate on the medium-access process under legacy and NPCA operation. Fig. 2 contrasts the two medium-access processes. Under legacy access (top), prior to IEEE 802.11bn, a STA contends only on the BSS primary channel.[1] It decrements its backoff counter while the channel is idle and transmits when the counter reaches zero. If the secondary channel is also sensed idle at this instant, the STA may bond it with the BSS primary channel and transmit a wideband PPDU spanning the combined bandwidth (e.g., 40, 80, or 160 MHz). When an overlapping BSS (OBSS) transmission seizes the BSS primary channel, however, the STA must defer and freeze its backoff for the entire OBSS busy period, even though the secondary channel is idle.

Under NPCA access (bottom), a STA no longer wastes the OBSS busy period. When it detects an OBSS transmission on the BSS primary channel, it decodes the preamble of that transmission and learns its duration in advance,[2] rather than freezing its backoff for the whole period, it switches to the idle NPCA primary channel and contends and transmits there. On each such switch the STA saves its backoff state on the BSS primary channel and initializes a fresh contention window for the NPCA primary channel, so every NPCA visit begins at the minimum backoff stage regardless of prior history. The STA already knows when the OBSS transmission will finish, so it sets a timer to that instant and switches back to the BSS primary channel when the timer expires, restoring the saved state without having to re-sense the channel. In this way the secondary spectrum that the legacy STA leaves idle is put to use, and a STA can complete one or more transmissions that it would otherwise have had to postpone.

Two constraints shape this access. First, a radio transition delay δ_{sw} is incurred on each switch. Second, the opportunity

[1] We use the term BSS primary channel to distinguish the original primary channel from the NPCA primary channel introduced in IEEE 802.11bn. Prior to 802.11bn, when NPCA was not yet defined, this channel was referred to simply as the primary channel.

[2] NPCA relies on preamble detection rather than energy detection: a STA must decode the inter-BSS PPDU preamble (L-SIG and BSS color) at PHY-RXSTART to both classify the transmission and derive its remaining duration, from which the NPCA window is determined. Energy detection alone, which yields no length or BSS information, is insufficient to trigger an NPCA transition.

lasts only until the known end of the OBSS transmission, whose remaining duration D_{rem} is read from the preamble. Contention and transmission must therefore fit within a bounded window of

$$W_{\text{eff}} \;=\; \left\lfloor \frac{D_{\text{rem}} - \delta_{\text{sw}}}{\sigma} \right\rfloor \quad \text{slots}, \tag{1}$$

where σ is the slot time, typically 9 μs in IEEE 802.11-compliance systems. This deadline is the crux of NPCA: every backoff decision competes against it. A STA that collides doubles its contention window as usual, and once the resulting backoff exceeds the slots left in W_{eff}, it can no longer transmit before the window closes so it wastes the rest of the opportunity despite having data to send.

Because the contention window is reset at every NPCA visit, this penalty recurs each time rather than amortizing across visits. This mismatch between open-ended backoff and the finite window is the inefficiency that PACE targets.

C. Slotted Contention Model

IEEE 802.11 STAs access the medium through the DCF, a CSMA/CA mechanism in which each STA maintains a backoff counter rather than transmitting at random in every slot. Directly analyzing DCF over a finite window is intractable, so we adopt the standard slotted abstraction introduced by Bianchi [**?**]: the backoff process of a saturated DCF STA is represented by a single per-slot transmission probability τ, the steady-state probability that the STA transmits in a randomly chosen slot. For the standard baseline this is purely an analytical model of DCF—the mechanism itself remains backoff-based—while τ also serves as the per-slot access-probability variable in which we express and compare the schemes studied here, including the proposed policy of Section IV.

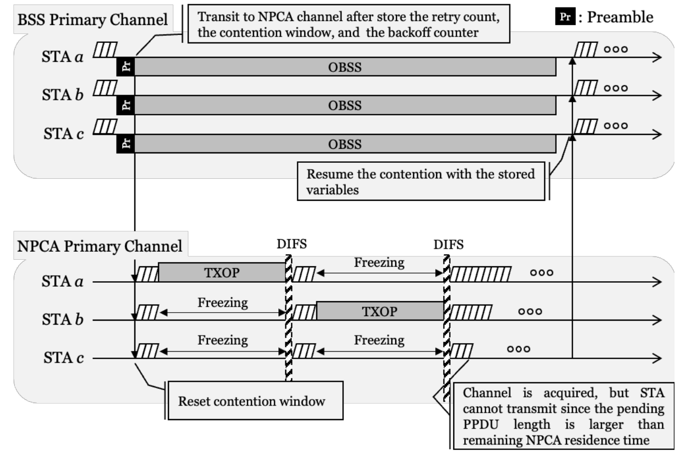
Under this abstraction, we model time on the NPCA primary channel as a sequence of slots of duration σ and index them by $t \in \{0, 1, \ldots, W_{\text{eff}}\}$. We denote by $W_{\text{rem}}(t) = W_{\text{eff}} - t$ the number of slots remaining at slot t, so that W_{rem} counts down toward the deadline as contention proceeds.

Let $\mathcal{V} \subseteq \mathcal{N}$ be the set of STAs that switch to the NPCA primary channel, each holding a pending PPDU whose transmission occupies L_i slots. Because a transmission must complete before the window closes, STA i is viable at slot t only if its frame still fits, i.e. $L_i \leq W_{\text{rem}}(t)$; otherwise it can no longer succeed and should remain silent to avoid wasting slots. We write the viable population at slot t as

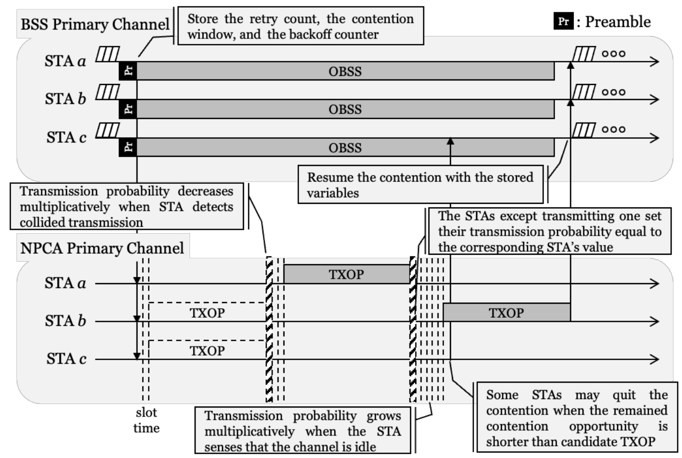
$$\mathcal{V}(t) \;=\; \left\{\, i \in \mathcal{V} : L_i \leq W_{\text{rem}}(t) \,\right\}, \tag{2}$$

which is non-increasing in t as both the window depletes and frames become infeasible.

At each slot, a viable STA i attempts transmission with probability τ_i. Following the half-duplex assumption, a transmitting STA cannot sense the channel in the same slot. Let $\mathcal{D}(t)$ denote the set of STAs whose transmissions on the NPCA primary channel are detected by a given STA in slot t. This set is defined at the channel level: it counts every STA transmitting on that channel in the slot, not only the switching visitors but



(a) NPCA baseline (binary exponential backoff).



(b) Proposed PACE (probabilistic adaptive contention).

Fig. 3: Medium access on the NPCA primary channel: a STA that detects an OBSS transmission on the BSS primary channel migrates and contends within the bounded NPCA window. (a) the standard NPCA baseline with binary exponential backoff, and (b) the proposed PACE with probabilistic adaptive contention and PPDU-aware self-exclusion.

also any other STA that uses the same channel as its own primary. The slot outcome is idle when $|\mathcal{D}(t)| = 0$, a success when $|\mathcal{D}(t)| = 1$, and a collision when $|\mathcal{D}(t)| \geq 2$. These three observable events are the only feedback available to a STA, and they drive the adaptive contention control developed in Section IV.

The remaining window advances by the time each outcome occupies the medium. A busy slot lasts until the longest frame it carries has ended, so it consumes

$$L_{\max}(t) \;=\; \max_{i \in \mathcal{D}(t)} L_i \tag{3}$$

slots: for a success this is the sole sender's length, and for a collision it is the longest colliding frame, as in standard IEEE 802.11 without collision detection. An idle slot consumes one slot. Charging a collision the full frame length is the conservative choice; a bounded-collision variant, in which an early collision detection or notification aborts the colliding frames after a short overhead, would only reduce this cost and is discussed in Section V.

Fig. 3 illustrates this slotted contention on the NPCA primary channel and contrasts the two schemes studied in this paper. In both, a STA that detects an OBSS transmission on the BSS primary channel stores its DCF state, migrates to the NPCA primary channel, and contends there until the NPCA timer expires, whereupon it restores the saved state and resumes on the BSS primary channel. Under the NPCA baseline of Fig. 3a, STAs contend with Binary Exponential Backoff: a STA that collides doubles its contention window, and one whose pending TXOP no longer fits the remaining window acquires the channel only to find it cannot transmit, wasting the slots it occupied. Under the proposed PACE scheme of Fig. 3b, each STA instead adapts its per-slot probability τ_i from the idle, success, and collision feedback—raising it multiplicatively on idle slots, lowering it on collisions, and copying a successful sender's value—while proactively withdrawing from contention once its frame can no longer fit within $W_{\text{rem}}(t)$. We detail this mechanism in Section IV.

IV. PACE Algorithm

Algorithm 1 summarizes PACE as executed independently by each STA on the NPCA primary channel during a single NPCA visit. Every STA runs the same rule using only locally observable feedback, those are, the per-slot outcome and its own remaining window without central coordination or knowledge of the contending population.

Algorithm 1 is expressed in the per-slot transmission-probability model of Section III-C: a viable STA transmits in a slot with probability τ_i. This is a probabilistic, slotted-ALOHA-style access policy.

We adopt the probabilistic approach for two reasons. First, the per-slot probability τ is the common abstraction under which DCF, PND, and AND are all analyzed [?], [?], [?], so it places PACE on the same analytical footing as the baselines we compare against. Second, and more fundamentally, the finite NPCA window is precisely where a deterministic countdown breaks down: a single backoff draw cannot re-adapt within the handful of slots a visit lasts, whereas a per-slot probability can be re-steered every slot by the feedback law, which will be described in Subsec. IV-D. The ALOHA-style attempt is therefore not incidental but the mechanism that makes within-window adaptation possible.

Algorithm 1 runs four phases per visit: (a) a one-time probability initialization, followed by a per-slot loop of (b) self-exclusion, (c) channel access, and (d) feedback-based adaptation. The remainder of this section develops each phase in turn.

A. Probability Initialization

A visitor that has just switched to the NPCA primary channel does not know how many other STAs are contending there. The population-matched choice $\tau_0 = 1/N$, which is the well-known throughput-optimal attempt probability of slotted ALOHA, is therefore unavailable in practice since N is not observable.

Algorithm 1 PACE at STA i during one NPCA visit

Require: PPDU length L_i (slots), window W_{eff}, factors $c_{\text{coll}}, c_{\text{idle}} > 1$

/* Phase (a): Probability Initialization */

$\tau_i \leftarrow 1/W_{\text{eff}}$; $W_{\text{rem}} \leftarrow W_{\text{eff}}$

while $W_{\text{rem}} > 0$ **do**

  /* Phase (b): PPDU-aware Self-Exclusion */

  if $L_i > W_{\text{rem}}$ **then**

    $\tau_i \leftarrow 0$

  end if

  /* Phase (c): Probabilistic Channel Access */

  draw $u \sim \text{Uniform}(0,1)$

  if $u < \tau_i$ **then**

    transmit the PPDU advertising τ_i

    if solo success **then**

      break

    end if

    $W_{\text{rem}} \leftarrow W_{\text{rem}} - L_{\max}(t)$ {collided: medium busy for the longest colliding frame}

  else

    /* Phase (d): Feedback-based Adaptation */

    observe the slot outcome $|\mathcal{D}(t)|$

    if $|\mathcal{D}(t)| = 0$ **then**

      $\tau_i \leftarrow \min(\tau_i \cdot c_{\text{idle}}, 1)$; $W_{\text{rem}} \leftarrow W_{\text{rem}} - 1$

    else if $|\mathcal{D}(t)| = 1$ **then**

      $\tau_i \leftarrow \tau_{\text{sender}}$; $W_{\text{rem}} \leftarrow W_{\text{rem}} - L_{\max}(t)$

    else

      $\tau_i \leftarrow \tau_i / c_{\text{coll}}$; $W_{\text{rem}} \leftarrow W_{\text{rem}} - L_{\max}(t)$

    end if

  end if

end while

Visitor STAs might, through association state or management traffic, form some estimate of how many STAs belong to its own BSS. The harder obstacle is that the population it actually contends with also includes the native STAs of the neighboring BSS that owns the NPCA primary channel. That is, a visitor competes against these as well, yet has no direct way to learn how many are active. The count that would parameterize $1/N$ is thus doubly hidden, which motivates a surrogate that avoids counting STAs altogether.

We propose a method where the window length itself determines the scale. A visit spans W_{eff} slots, and a STA contends in at most one of them per slot so W_{eff} upper-bounds the number of attempt opportunities any STA sees in a visit. A STA attempting with probability τ therefore expects at most $W_{\text{eff}}\,\tau$ attempts before the deadline. Fixing this budget to a single probe per STA,

$$W_{\text{eff}}\,\tau_0 = 1, \tag{4}$$

sets the most conservative self-consistent exploration rate: low enough to avoid an early collision storm, yet high enough for each STA to test the medium about once before the window drains. This yields

$$\tau_0 = \frac{1}{W_{\text{eff}}}, \tag{5}$$

which depends only on the effective window W_{eff}, which is a quantity that each STA can compute locally from the OBSS PPDU duration. No knowledge of the contender count is required.

In general frame length to be transmitted on NPCA primacy channel is longer than a single slot, so τ_0 is bound to be smaller than the optimal value. This choice is deliberately biased toward under-attempting for two reasons. First, it is self-correcting in the safe direction: an over-conservative τ_i is raised toward the operating point within a few slots by the idle-driven multiplicative increase in Phase (d), whereas an over-aggressive start triggers collisions whose wasted slots cannot be reclaimed inside a finite window. The MIMD recovery is thus asymmetric, and starting low pays only a bounded ramp-up cost. Second, $1/W_{\text{eff}}$ tracks the regime where initialization matters most: when the window is tight ($W_{\text{eff}} \approx N$) it already approximates the optimum $1/N$, and when the window is loose ($W_{\text{eff}} \gg N$) the mild initial timidity is erased by the idle increase long before the window drains.

B. PPDU-aware Self-Exclusion

At the start of every slot a STA compares its own frame length L_i to the remaining window $W_{\text{rem}}(t)$. If $L_i > W_{\text{rem}}(t)$ the frame can no longer finish before the deadline, so the STA sets $\tau_i = 0$ and falls silent for the rest of the visit. The test is purely local. That means it involves only the STA's own L_i and the slot index and thus, like initialization, requires no knowledge of the contending population.

Self-exclusion keeps the optimal target that PACE aims to track,

$$\tau^*(t) \;=\; \frac{1}{|\mathcal{V}(t)|}, \tag{6}$$

meaningful in a heterogeneous and finite window. In a slot with $|\mathcal{V}(t)|$ viable STAs, a success, meaning exactly one transmitter, is most likely when each attempts with probability $\tau^*(t)$, and this target rises over the visit as the viable set drains. A STA whose frame no longer fits is removed from the viable set $\mathcal{V}(t)$ of Eq. (2), so it neither wastes time on attempts destined to fail nor inflates the contention seen by the STAs that can still succeed. This plays the same role as the collision detection departure of the original neighbor discovery algorithm, in which a STA leaves the contention once its advertisement succeeds. Here a STA also leaves once its frame provably cannot complete, tightening $\mathcal{V}(t)$ toward the set of STAs that genuinely compete for the remaining slots.

C. Probabilistic Channel Access

A STA that is still viable transmits in the current slot with probability τ_i and otherwise listens. Because the radio is half-duplex, a transmitting STA cannot sense the slot it occupies, so it performs no adaptation on its own transmissions. Each transmitted PPDU advertises the sender's current τ_i in its header, which the listeners use in Phase (d).

A slot is a success when exactly one viable STA transmits and its frame fits the remaining window. The winner then departs the visit—mirroring the post-success departure of neighbor discovery—which decrements $|\mathcal{V}(t)|$ and raises the optimal target $\tau^*(t)$ for the STAs that remain. Slots with no transmission are idle, and slots with two or more transmissions are collisions; both outcomes are observed by every non-transmitting STA and feed the adaptation of Phase (d).

D. Feedback-based Adaptation

Every viable STA that listened in the slot updates τ_i from the slot outcome $|\mathcal{D}(t)|$ using the multiplicative-increase and multiplicative-decrease (MIMD) rule inherited from neighbor discovery as

$$\tau_i \leftarrow \begin{cases} \min(c_{\text{idle}}\,\tau_i,\,1) & |\mathcal{D}(t)| = 0 \quad \text{(idle)}, \\ \tau_{\text{sender}} & |\mathcal{D}(t)| = 1 \quad \text{(solo)}, \\ \tau_i / c_{\text{coll}} & |\mathcal{D}(t)| \geq 2 \quad \text{(collision)}, \end{cases} \tag{7}$$

with $c_{\text{idle}}, c_{\text{coll}} > 1$.

An idle slot signals that the population has been overestimated and the attempt rate is raised by the factor of c_{idle}. On the other hand, a collision signals the opposite, hence, once STAs detect a collision, the attempt rate is divided by c_{coll} to reduce the contention. On a solo success, every listener copies the advertised τ_{sender}, so a single successful slot propagates the winning operating point to the whole contending set at once, giving PACE the fast intra-visit consensus.

None of the three updates counts the viable STAs. Instead, the idle and collision statistics move each τ_i toward the operating point implicitly, and solo-copy pulls the population onto a common value, so the ensemble approximately tracks the time-varying optimum $\tau^*(t)$ of Eq. (6) without estimating $|\mathcal{V}(t)|$.

The direction of this adaptation is what separates PACE from legacy Binary Exponential Backoff. Because the viable set only shrinks over a visit—STAs depart on success or self-exclude when their frame no longer fits—the optimal target $\tau^*(t) = 1/|\mathcal{V}(t)|$ of (6) rises as the deadline approaches. BEB moves the opposite way: a collision doubles the contention window and halves the attempt rate, so the access probability falls exactly when it should climb. The mismatch is twofold. First, the adaptation is backwards—BEB retreats on congestion while the window's diminishing population calls for a more aggressive attempt rate—so late slots are left idle. Second, the doubled backoff can exceed the remaining window $W_{\text{rem}}(t)$, silencing the STA for the rest of the visit and wasting those slots outright; unlike PACE's self-exclusion, which silences only STAs that provably cannot finish, this dead time falls on STAs that could still have succeeded. PACE's idle-increase and solo-copy instead move τ_i along the rising $\tau^*(t)$ trajectory, matching the sign of the optimal policy rather than fighting it.

TABLE II: Simulation Parameters

Parameter	Value
Contending STAs N	$\{10, 20, 30, 50\}$
Effective window W_{eff}	$\{20, 50, 100, 200\}$ slots
Slot time σ	9 μs
PPDU length (homo / uniform / bimodal)	6 / $\mathcal{U}[3, 12]$ / $\{4, 12\}$ slots
Native STAs M (mixed setting)	$\{0, 5, 10, 20\}$
Initial probability τ_0	$1/W_{\text{eff}}$
MIMD factors $c_{\text{coll}}, c_{\text{idle}}$	1.2
DCF window CW_0 / $\text{CW}_{\max}$	N / 1023
Collision cost	$\max_{i \in \mathcal{D}(t)} L_i$
Visits per point / seeds	1000 / 5

E. Coexistence with Native STAs

The NPCA primary channel is not unused spectrum reserved for the target BSS: Channel #2 is itself the BSS primary channel of a neighboring BSS, comprising its own AP and a set $\mathcal{M} = \{1, 2, \ldots, M\}$ of native STAs that contend on it with standard DCF/EDCA at all times. The visitors, by contrast, use Channel #2 only opportunistically, during NPCA visits. Two contention disciplines therefore coexist on the NPCA primary channel: the native STAs' persistent DCF and the visitors' bounded PACE-driven access.

Because the slot outcome $\mathcal{D}(t)$ is measured on the channel, native activity enters a visitor's feedback directly. A slot in which two or more frames overlap is a collision whether the extra transmitter is a visitor or a native, so a native transmission that clashes with a visitor lowers that visitor's τ_i through (7), and a slot with no transmission at all, native or visitor, is idle and raises it. The solo case is the one exception. The copy $\tau_i \leftarrow \tau_{\text{sender}}$ requires the successful frame to advertise a probability, which only a PACE visitor does. When a native transmits alone its frame carries no such value, so a listening visitor keeps τ_i unchanged and simply observes the busy slot while the window continues to shrink.

Because the two BSSs only partially overlap in coverage (Fig. 1), the native contention a visitor experiences on the NPCA primary channel depends on its location. A visitor inside the overlap region senses, defers to, and competes with the native STAs, whereas a visitor outside the neighboring BSS coverage encounters no native traffic there and effectively reuses the channel without contending with natives. The co-existence cost to the neighboring BSS is thus greatest when every visitor lies within its coverage. Unless otherwise stated we adopt this full-overlap configuration—a single collision domain in which all visitors and native STAs are mutually within carrier-sensing range—since it upper-bounds the impact on native airtime; the partial-overlap regime, in which non-overlapping visitors enjoy interference-free spatial reuse, is left to future work. The resulting impact on native airtime under full overlap is evaluated in Section V.

V. Performance Evaluation

We evaluate PACE in the slotted NPCA contention model of Section III-C; Table II summarizes the parameters. Unless stated otherwise, a visit is contended by $N \in \{10, 20, 30, 50\}$ stations over an effective window $W_{\text{eff}} \in \{20, 50, 100, 200\}$ slots, PPDU lengths are drawn from a homogeneous, uniform $\mathcal{U}[3, 12]$, or bimodal $\{4, 12\}$ distribution, and each data point averages 1000 visits over five seeds. PACE uses $\tau_0 = 1/W_{\text{eff}}$ and $c_{\text{coll}} = c_{\text{idle}} = 1.2$. A collision is charged the full length of the longest colliding frame, eq. (3). We compare PACE against standard DCF with binary exponential backoff, the open-loop AND schedule [?], and an oracle that transmits at $\tau^*(t) = 1/|\mathcal{V}(t)|$ with perfect knowledge of the viable set. Performance is reported as the window utilization $\sum_i L_i \, \mathbf{1}[\text{success}]/W_{\text{eff}}$, its ratio to the oracle (efficiency), and, for fairness, Jain's index and a native-versus-visitor proportionality index.

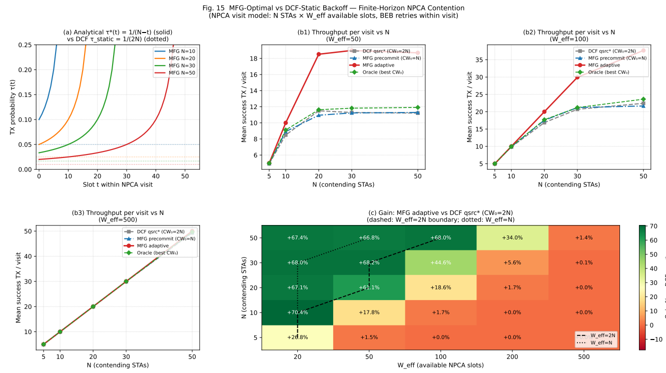


Fig. 4: Window utilization of the throughput-optimal policy and DCF versus window tightness.

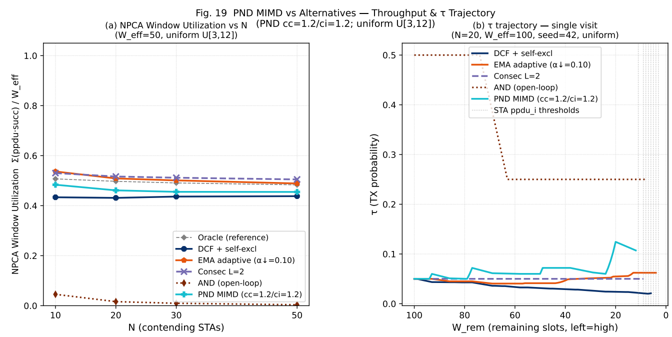


Fig. 5: Per-slot attempt probability over a visit and window utilization versus N, for PACE, DCF, AND, and the oracle.

A. Utilization Gain in the Tight-Window Regime

Fig. 4 isolates the regime in which adaptive access matters. As the window tightens toward $W_{\text{eff}} \approx N$, the fixed backoff of DCF leaves a growing share of the window idle, and the gap to the time-varying optimum widens to over 70%; once $W_{\text{eff}} \gg N$ the window is no longer the binding constraint and DCF recovers. This identifies the tight window as the operating point that the rest of the evaluation targets.

B. Tracking the Time-Varying Optimum

Fig. 5 shows why PACE closes this gap. Over a single visit its ensemble attempt probability rises along the oracle's $\tau^*(t)$ track as the viable set drains, whereas DCF sawtooths in the opposite direction, halving its rate on every collision, and AND

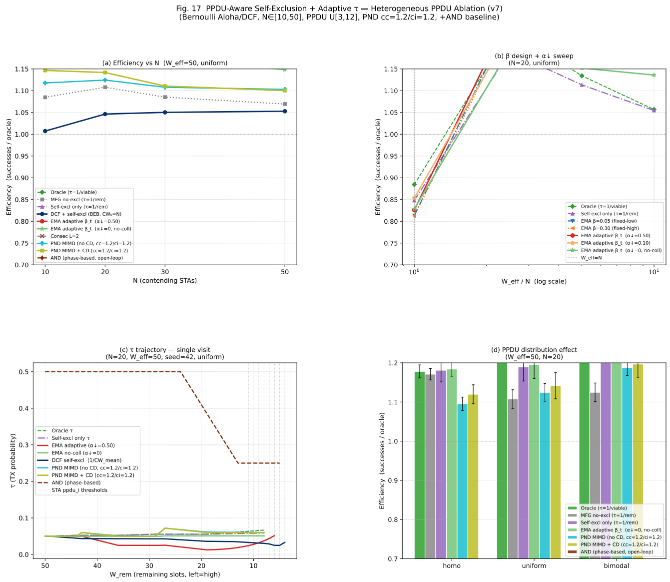


Fig. 6: Efficiency relative to the oracle under heterogeneous PPDU lengths, with and without PPDU-aware self-exclusion.

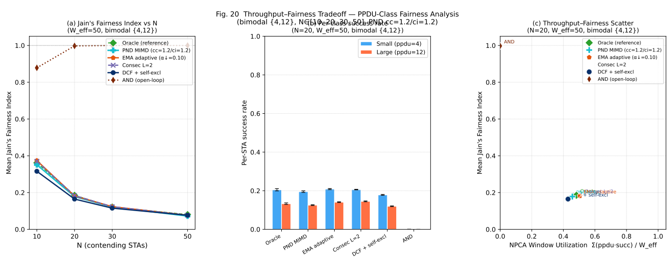


Fig. 7: Aggregate throughput versus Jain's fairness index across PPDU classes for PACE, DCF, and AND.

follows a fixed open-loop schedule blind to the deadline. The resulting utilization keeps PACE close to the oracle across N, ahead of DCF, while AND collapses in the finite window.

C. Self-Exclusion under Heterogeneous Frame Lengths

Fig. 6 isolates the contribution of the self-exclusion rule. When frame lengths are heterogeneous, silencing stations whose frame no longer fits the remaining window recovers the slots they would otherwise waste on doomed attempts, lifting efficiency toward the oracle; adding the MIMD feedback on top closes most of the residual gap. The ordering is preserved across the PPDU distributions and confirms that both components of PACE are needed.

D. Throughput–Fairness Trade-off

Fig. 7 examines whether the utilization gain comes at the expense of fairness across frame-length classes. PACE Pareto-dominates DCF, achieving both higher aggregate throughput and a higher Jain index, while AND sacrifices throughput almost entirely. PACE thus improves efficiency without trading away inter-class fairness.

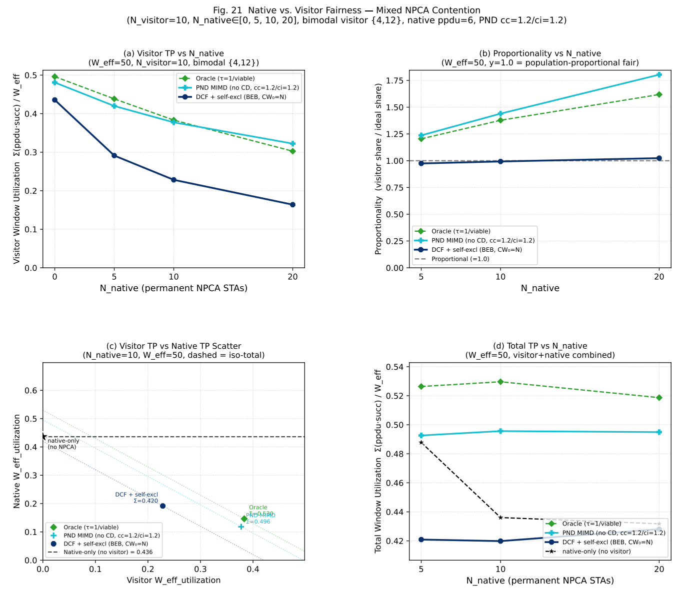


Fig. 8: Visitor and native window utilization when NPCA visitors share the channel with permanent native DCF stations.

E. Coexistence with Native Stations

Fig. 8 evaluates the mixed setting in which NPCA visitors contend against the native stations that own the channel. PACE raises visitor throughput and total channel utilization well above DCF, recovering idle airtime that would otherwise be lost. This gain does come at a cost in native airtime, which we report rather than hide: because PACE is more aggressive than the collision-averse DCF, it preserves less of the native throughput, a trade-off we return to in the discussion.

F. Robustness to the Collision-Cost Model

Finally, Fig. 9 tests whether the advantage of PACE is an artifact of the collision-cost assumption. Charging a collision one slot, as an ideal collision detection would, amplifies the gain of PACE over DCF; charging it the full frame length, as standard IEEE 802.11 does, reduces every scheme's utilization but preserves the ordering, with PACE still ahead of DCF. The benefit of PACE therefore holds across the full range of collision-cost models, and is only enhanced when a bounded-collision mechanism such as CSMA/CN is available [**?**].

VI. Conclusion

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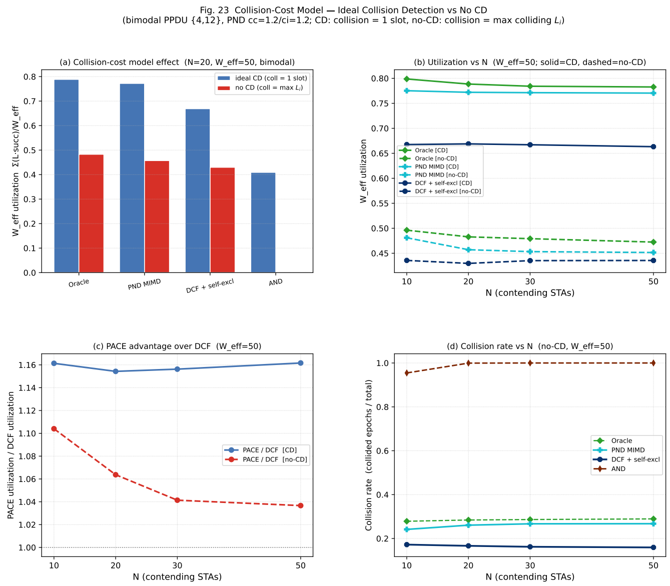

Fig. 9: PACE and DCF utilization under an ideal collision detection model (collision costs one slot) and the no-detection model (collision costs the longest colliding frame).

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